

Livestreaming a *doitocracy*: Platform-jumping participatory practices in modular synthesis gear cultures

Convergence: The International
Journal of Research into
New Media Technologies
2026, Vol. 32(2) 561–577
© The Author(s) 2026

Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/13548565261417339
journals.sagepub.com/home/con



Eliot Bates¹ 

Abstract

Earth Modular Society (EMS) is a cross-platform community dedicated to live hardware modular synthesis, with their primary efforts going into a 24/7 modular “radio station” that livestreams on Twitch and YouTube, and a supporting Discord chat server. Although part of the broader modular synthesis gear culture (a large-scale social formation that coalesces around specific classes of fetishized technical objects), EMS represents a new development in gear cultures as it foregrounds live performance and minimizes the accrual of status via conspicuous consumption. The normative governance structures of platform-specific gear culture communities preclude all but one or a handful of users from having any meaningful input into the rules, governance, or participatory modes of the platform. In contrast, EMS is structured as a *doitocracy*, where all are encouraged to bring things to do to the community, and to support each other in doing. The *doitocracy* concept contributes to understanding the wide range of DIY (do-it-yourself) and DIT (do-it-together) community practices. This *doitocracy* case study also demonstrates the importance of analyzing the performance of crafted objects, and how these, alongside platform-jumping practices, can constitute primary organizational forces in online/offline community sociality.

Keywords

audio equipment, *doitocracy*, electronic music, gear, gear cultures, livestreaming, modular synthesis, music technology, platform governance, platform-jumping, synthesizers

¹Music Department, CUNY Graduate Center, New York, NY, USA

Corresponding author:

Eliot Bates, Music Department, CUNY Graduate Center, 365 Fifth Ave, 3rd Floor, New York, NY 10016, USA.

Email: ebates@gc.cuny.edu

Introduction

Any day of the week, at any time, if you want to hear music made with hardware modular synthesizers, you can tune into Earth Modular Society's Twitch or YouTube channels to catch their nonstop livestream. The sounds you hear are never pre-recorded or "AI" generated; they are streamed live from *someone's* synthesizer. Videos are broadcast from around the world from many kinds of spaces: home or professional recording studios, bedrooms, living rooms, back porches, or live music venues. The society formed during the COVID-19 lockdowns, but continued long after other music communities phased out livestreamed events and returned to promoting in-person shows. There are no ads, no sponsors, nothing is monetized, there usually is no schedule, and no one person is obviously responsible for the massive effort. Yet the livestreams, and the more than 16,400 performance videos that have been clipped from those streams and archived to YouTube, today represent the largest and most eclectic repository of amateur modular synthesis performances anywhere. The group has a distinctive approach to participation, what they call a *doitocracy*. This *doitocracy* is a kind of DIY/DIT practice (do-it-yourself/do-it-together) applied towards running a 24/7 modular synthesis radio station, but is expanded to encompass informal online pedagogy for synthesizing and the collaborative design of new products for the synthesizer market. What is the *doitocracy*? How does it relate to the normative means of producing and consuming synthesizer music? And how do "just-do-it" community practices differ from other music-technology-centric sociocultural formations?

Earth Modular Society is an example of a recently formed online community that was formed through the convergence of several media that are normally regarded and analyzed as distinct: community radio, livestreaming, chat, podcasting, and local in-person open mic performances. But, contra Tia DeNora's provocation that "music has organizational properties" (2000: 151), in this instance it is musical equipment—gear—that is entrusted to having such organizational properties. The lack of an overarching musical genre (Beer, 2013), and therefore an analyzable style, for the livestreams (which have included genres as disparate as ambient, noise, techno, drum and bass, experimental, drone, sound art, synthwave, and academic electroacoustic music) immediately sets this community apart from the many scenes and subcultures analyzed in the literature of popular music studies and media studies (e.g., McRobbie, 1980; Straw, 2001). Therefore, this article argues that sociality should be studied via the platform-jumping practices within predominantly online communities, and that the mediatized (re)presentations of human-material encounters—gear performances and discourses—should be understood as constitutive of online sociality and community-building. Moreover, the *doitocratic* emphasis on "just doing it" over conspicuous consumption sets Earth Modular Society apart from previously studied gear cultures (Bates and Bennett, 2022).

This research builds on 9 years of mixed-mode, multi-sited research into modular synthesis gear cultures, utilizing a gear cultures methodology (Bates, 2026a). This consists of five interwoven approaches and access points:

- (1) Conducting a reception ontology of **gear** to produce a material account built upon the fullest known set of meanings, practices, and attitudes towards gear
- (2) Using sensory ethnographic methods to investigate the subjectivization and individualization processes that configure gear **users**
- (3) Tracing power within **social** relations, especially around fetishization practices
- (4) Utilizing anthropological participant observation to study these novel social formations as **cultures**, complete with their own distinctive histories, mythologies, argot, jokes, visual and material culture, and values
- (5) Employing targeted theoretical sampling to identify **limit**/deviant cases to be analyzed via the four preceding approaches, and to test the bounds of the gear culture.

I initially focused on circuit designer communities before I more widely considered the online/offline social formations around synthesizer gear. Here, I focus on the fifth approach/access point, and discuss one synthesis collective that serves as the primary limit case in the broader study. In addition to participant observation in Twitch/YouTube livestreams and in one Discord server, and taking fieldnotes at the EMS trade-show booth at the Knobcon trade-show festival in Chicago in 2023, I conducted subsequent focus group interviews with five active admins/moderators (all of whom are regular participants and performers as well).¹ Since my focus here is not primarily on the artistic material or performance of modular synthesis, readers can gain an aesthetic sense of the main public-facing representation of EMS via the archive of more than 16,400 performer streams on EMS' YouTube channel.²

First, I will outline some basic concepts specific to modular synthesizers and the history of social interaction around them. Then, I will discuss how modular synthesizer sociability transitioned from being a collection of small-scale local scenes to becoming a transnational *gear culture* (and what that means). The online interactions of modular gear cultures, unlike those of other gear cultures, normally coalesce around Discord chats, livestreaming, and YouTube content creation. I examine the governance structure of and nature of participation within the many creator-centered communities and regional synth societies. The remainder considers one case study—EMS (Earth Modular Society)—and the nature of their “just do it” ethos.

Hardware modular synthesizers: From solo practice to gear culture

Modular synthesizers are collections of discrete objects, called *modules*, that their users connect in novel ways to produce unique, bespoke musical instruments. Each module represents one building block of synthesis, for example, audio samplers, logic operators, tactile controllers, oscillators as waveform generators, filters to change their timbre, envelopes or function generators for creating a specific progression of voltages over time or to add shape to sounds, and voltage controlled amplifiers for allowing the loudness of a signal to be adjusted by a changing voltage (Navs, 2016). In hardware modular synthesis, and skeuomorphically rendered in software modular, connections between and within modules are made with patch cables. Accordingly, “a patch” represents one unique configuration that may contain anywhere from twenty to hundreds of cable connections between three and dozens of discrete modules (Figure 1). A popular narrative of technological progress, instigated by magazine writers but adopted by academics, has it that modulars were replaced in the 1970s by all-in-one keyboards, which were replaced in the 1980s by digital devices, with an “analog revival” of sorts starting in the early 2000s (Pinch and Trocco, 2002: 317–320; Nelson et al., 2023). This is misleading. While never a top-selling product category, analog modular synths have been continuously made as boutique products, and played by musicians, ever since the first Moog and Buchla instruments circulated in 1965. Such claims also ignore the empirical problem that most post-2000 hardware modular synths contain both analog and digital modules.³ Some modular performers today lack any nostalgia for a golden era of analog, viewing modular as a space for tactilely exploring computation and mathematics in music (Navs 2016).

Modular synthesis appears at first to be a predominantly solo affair, a solo practice characterized by the sole ownership of a collection of objects, to be played solo. Prominent synthesists from the 1960s and 70s, such as Wendy Carlos and Suzanne Ciani and Delia Derbyshire, have largely been portrayed in academic and journalistic literature as artists laboring by themselves in isolated recording studios, immersed in their own world of sounds typically inaccessible to others (Morgan, 2017; Pinch and Trocco, 2002: 131–170; Rogers, 2020). The main exception to this were university electroacoustic music studios (Schedel, 2017) which were, for many years, among the only

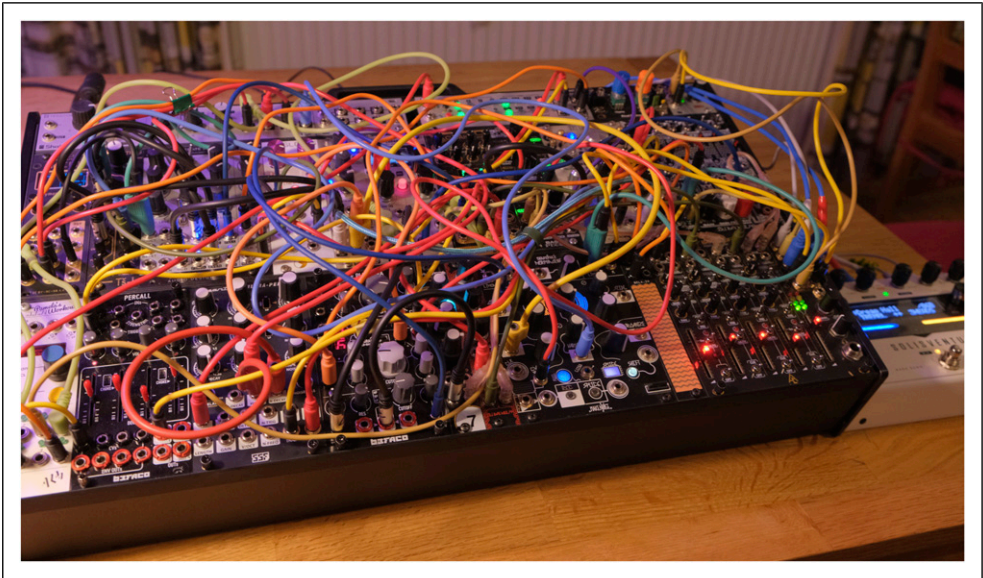


Figure 1. The author's modular synthesizer, July 2025. Photo by the author.

institutions where multiple people could gain access to the same instruments. But even there, these instruments were infrequently used within collaborative, collective musical practices.

Online modular-related social interaction began around 1995 with synth-diy, a mailing list populated by a few analog synthesis enthusiasts who were trying to figure out how to make and repair instruments. The establishment and adoption of several open hardware formats, especially the Eurorack format, led *multiple* designers to create modules for the *same* format—a significant departure from the proprietary, single-manufacturer systems of Moog, Buchla, Serge, and ARP (Teboul, 2024). This, alongside the distribution of home-printed analog synthesis circuit “cook-books” (Henry, 2008; Hutchins, 1972–2004; Wilson, 2013), and an emerging online culture of sharing schematics, techniques, components, and personal contacts at component firms, helped hardware modular to become a creative and social space for musical circuit and interface design (Kitzmann and Engström, 2024).

For those uninterested in becoming circuit designers, the main modular synthesis social opportunity, which gained steam in the 2000s, was the neighborhood meetup: intrinsically local affairs where individuals would bring their modulators, set them up on tables, and stand by in case anyone wanted to ask questions about the objects. At one I attended in Ithaca, New York in 2010, a half dozen instruments were all plugged in and set up, but the owners never touched them. Instead, the modulators were patched to autonomously produce generative sequences of pitches and modulations: plug it in, and it would run indefinitely, the sound subtly changing.⁴ At some neighborhood meetups the instruments were not connected to speakers at all: we only saw the instruments, blinking lights ablaze. Rarely did more than a couple dozen enthusiasts attend a neighborhood meetup, and often the same (small) crowd would show up, making some early modular events seem akin to music scenes (Straw, 2001).

However, in the 2010s, two widespread phenomena changed modular's social form into a novel kind of social formation known as a *gear culture* (Bates and Bennett, 2025). In disparate locations in the US and Europe (e.g., Frankfurt and Berlin, DE; Schaumburg, IL; Brooklyn, NY; Barcelona, ES;

Leeds, UK) organizers experimented in creating a novel kind of event, what I call a *trade-show festival*, which featured dozens or hundreds of exhibitor booths staffed by synth designers, a DIY area for synth-making workshops, talks by designers, some form of performance space, and one or more people documenting the event to share on YouTube and blogs. Alongside this, electronic musicians began uploading demonstrational audiovisual content to YouTube, and by the end of the 2010s, gear demos, gear reviews and gear tutorials were being consumed orders of magnitude more than any music that was produced with the gear. forestine, a modular society Discord admin and Twitch moderator (who we will meet again later), cogently described to me the space of online modular videos in late 2019, just prior to the first COVID-19 lockdowns in North America: “I was just getting into modular, and I kept trying to find people using it. And all I could find was videos where people were talking about the new product, like sponsored YouTube content.”⁵ The stars of the show appeared to be the synthesizers (or their modules) themselves, rather than musicians using them for musical purposes, or the music for that matter. And the two obvious pathways to becoming known and respected within this emergent gear culture were to either design and exhibit synths, or to produce content about them.

Gear cultures are novel social formations that coalesce following the confluence of several phenomena. A certain category of technical objects, known collectively as gear, become fetishized as magical objects (Pels, 2023). Those objects (more than the results of the normative activities done with them) become regarded as active agents in social relations. Novel forms of connoisseurship and valuation develop, and some “power users” begin to improvise full-time careers that provide products and services to the gear culture. Finally, the collective infatuation with a category of gear becomes widespread enough to encompass multiple dedicated internet platforms and transnationally held gear-specific events (Bates and Bennett, 2025). Largely discrete gear cultures have been identified around recording studio gear, hardware modular synthesizers, guitar pedals, DSLR/mirrorless photo equipment, film cameras, video equipment, espresso machines, overclocked gaming computers, and mechanical keyboards—but not all have been studied to the same extent.⁶ Several of these gear cultures formed when a category of technology became nominally obsolete, having allegedly been replaced by more affordable software simulations or mass-produced products. In opposition to this purported obsolescence, passionate users rejuvenated interest in the technology type and developed new practices and meanings around it.⁷ Beyond just representing a widespread, transnational, low-ticket luxury good market and consumption pattern for the contemporary equivalent of Veblen goods (Baudrillard, 1998; Veblen, 1899), gear cultures can become the primary social milieu for many thousands of participants—potentially 250,000 for the modular gear culture alone—some of whom strive to make a full-time living providing goods and services to the culture (Bates, 2026b).

DIY, modular, and community governance

While many aspects of modular’s social life are hyper-commoditized and individualistic, three aspects of modular synthesis clearly embody DIY/DIT culture. First is the aforementioned sharing culture among instrument designers, which buttresses a general sentiment that if you really want a particular kind of module to do a particular thing in a particular way, you should probably do it yourself.⁸ Second is the preponderance of DIY soldering and module-assembly workshops, either attached to trade-show festivals, or sometimes happening as regular regional events. Both are familiar to anyone who has encountered makerspaces or the maker movement (Davies, 2017; Hepp and Schmitz, 2022; Wasielewski, 2017); they are specialty forms of making and craft community practices. Third, and most recent, comes in the form of two different hardware synthesis online

communities, EMS (Earth Modular Society), and GSG (Golden Shrimp Guild). Both are focused on collaboratively produced livestreamed events, where members regularly interact around those and other topics in Discord servers. Their membership overlaps somewhat, and since a performer can stream to multiple channels and platforms simultaneously, some modular performances air on both. I will focus on EMS for the remainder of the essay for the sake of clarity, because EMS from the beginning has been explicit about educating their members in a “just do it” ethos, and because of the strong synergy between the community admins’ theoretical moves and what the community has become.

The development of collaborative performance broadcasting is novel within synthesis gear cultures, since almost every synthesis-related community or synthesis event has been the initiative of one or a very small number of people who subsequently maintain tight control over it. There are few opportunities, therefore, for widespread collective participation in determining what the nature of the community and events would or could be. Discord is a very popular chat platform for synthesis-related sociability, and the most popular synthesis Discord servers are centered on a single professional YouTube content creator, who offers Discord community access as a paid benefit for their Patreon subscribers. The general assumption appears to be that the fans of a particular creator, on account of having *one* shared interest, likely have *other* things in common that they could talk about. There may be additional incentives for joining; some creators solicit ideas or crowdsource samples for upcoming videos, or run competitions, or host monthly listening parties of music created by subscribers (as Red Means Recording and Benn Jordan do).

While there may be a *sense* of community participation in creator-specific communities, in reality all users are subject to the whims of one individual who makes a living by producing monetizable content to promote to the very people who are their Patreon subscribers. Thus, in trying to find community in an add-on perk to a Matryoshka doll of consumer product promotion, unsurprisingly, capital and capitalist relations enframe almost every aspect of interaction. It is as if Mark Fisher’s ruminations on depression and late capitalism, where “capitalism is what is left when beliefs have collapsed at the level of ritual or symbolic elaboration, and all that is left is the consumer-spectator, trudging through the ruins and the relics” (2009: 4), had become a community design principle. And while the entity in question differed, this immediately evoked Wendy Brown’s open question about governance and self-interest:

How does support for governance modeled on the firm and a normative social fabric of self-interest marry or jostle against support for governance modeled on church authority and a normative social fabric of self-sacrifice and long-term filial loyalty, the very fabric shredded by unbridled capitalism? (Brown, 2006: 692).

On these platforms, the most common discussion topics are about the buying and selling of synth-related objects, or the buying and selling of objects related to other hobbies and collections. The kind of capital invoked in this discussion, therefore, resembles the “audio capital” that is produced through the professional *use* of gear (O’Grady 2021), but unlike audio capital, can be accrued even if not used in music-making. Rather, discussions encompass the prices of new and used market commodities, the ethics of companies trying to undercut others by offering a product for cheaper, and money management tips such as strategies to help convince “the wife” that a purchase was justifiable.⁹ While these communities do have a life of their own to the extent that other topics unrelated to synthesis may be discussed, for example, internet memes and birdwatching on Red Means Recording’s Discord, parenting and wellness on DivKid’s Discord, in these cultural formations, physical objects—gear—ultimately frame *all* social relations.

The online platforms of the several dozen regional synthesis societies are similar to those around YouTube content creators in respects. They may be run by only one person, or by a self-appointed handful of people who were never elected. In all that I have participated in or observed, gear appears to mediate everything, and the in-person events these groups organize are typically more structured around the technological objects—the gear—than around music-making, music-listening, or shared sociopolitical values (Bates, 2026b). Hence, I do not try to shoehorn these communities into extant theories of scenes or subcultures.

This was the situation until late 2020: every trade-show festival, regional society, or YouTube content creator-specific community lacked any form of democratic or consensus governance. All were effectively monarchies, perhaps supported by a small aristocracy, where the monarch ruled by whim and could arbitrarily make and break rules. It is unclear why this was this way. Many dance-focused electronic music communities run as communes and collectives (St John, 2004). But, recall, modular communities were never primarily music communities; that difference became underscored in governance.

The lack of collective platform governance becomes especially visible when users post offensive content. The three largest online platforms for synthesis discussion (measured by registered users and number of posts)—the r/modular subreddit, the largest audio gear forum ([gearspace.com](https://www.gearspace.com)), and the largest dedicated modular synthesis message forum ([modwiggler.com](https://www.modwiggler.com))—maintain a long history of combative posts perceived by many to be misogynistic, homophobic, transphobic, and/or racist. This echoes other manosphere platforms (Bates and Bennett, 2022; Ging, 2017; Troitski and Bates, 2024). But as I have observed unfold in real time, and which is apparent on the admin/complaints subforums in moderator-user interactions, when a user complains about a post they find offensive, any subsequent action is subject to the whimsy of a moderator or site founder who was never chosen by the community, who arbitrarily interprets a set of rules that they can arbitrarily change at will—even retroactively. For example, Modwiggler and Gearspace maintained the sexualized and controversial names Muffwiggler and Gearslut for 15 and 19 years respectively, despite numerous name-change petitions and requests. The irony is that while a contingent of regular users frequently post about their right to “free speech,” ultimately they have no say into the site’s identity, its rules, its ethos, or its norms. Unsurprisingly, such platforms don’t work equally well for all participants; the love of gear can wear off. During my field research, I’ve had the opportunity to informally interview around two dozen “leavers,” people who left some specific online gear platform (other than EMS) and the surrounding gear culture. Typically, the circumstances were sufficiently traumatic that they quit that hobby altogether. Thus, there has long been the case for a fundamentally different kind of platform organized around different values.

The creation of EMS

During one COVID-19 lockdown in 2020, an English musician known as schmooster found himself online “eighteen hours a day,” participating in various online communities including Colin Bender’s Modular Mayhem, NYMS (New York Modular Society), and Look Mum No Computer’s message forum. They had a vision for a twenty-four-hour modular synthesis radio station, and began streaming their own synthesizer churning away at a generative patch for hours on end. Discovering that many of the existing communities either provided no performance opportunities or were very “gatekeepy,” schmooster began advertising their personal Twitch stream and Discord server (later renamed to Earth Modular Society, hereafter EMS) on these communities and even on LinkedIn. One example of the “gatekeepy” field of practice was the considerable financial outlay required to be a performer, where physical instruments routinely cost \$5000 to \$20,000. Free or low-cost software

modular instruments, and hybrids between software and hardware systems, were plentiful, but were typically prohibited by all existing online synth platforms and by the regional synth societies.

One of the first to join was an experimental ambient artist from the Canadian prairies named forestine, who had been helping NYMS with running their regular modular livestreams (Bates, 2022). They had previously played in a punk band but afterward “didn’t play music for about a decade because I felt gatekept from music.” Despite not having performed synthesizers live before, “modular allowed me to do stuff by myself without the permission of other people.” forestine was the first of a couple dozen modular enthusiasts who immediately saw potential for EMS, and set to expanding the range of Discord channels and promoting the various impromptu livestreams that members would broadcast on YouTube or Twitch (Figure 2). After many experiments with manually coordinating the activities of multiple streams, two users figured out server-side software scripts that would allow “hold music,” basically a continuous stream of sound coming live from some member’s unsupervised modular, to be “interrupted” by any streamer with their own stream, which upon ending would automatically revert to the hold music.

Having been one of the producers of NYMS’ livestreamed festivals and in-person concerts since 2020, early on I was familiar with EMS, and I even participated as a performer in three of their epic day-long “stream trains,” chock full of back-to-back performances spanning many electronic music genres and beyond. Like many modular enthusiasts, I often will be in the middle of work and get a notice that someone’s upcoming EMS stream is going live; I’ll tune in to catch it while working on other things (such as journal articles!) and occasionally post an encouraging comment to the performer on Twitch. The EMS Discord had a different vibe than any other to which I was subscribed. But, not knowing how that came to be, and not wanting my research to become dominated by auto-ethnographic accounts, I approached several of the nearly two dozen “admins”—members who

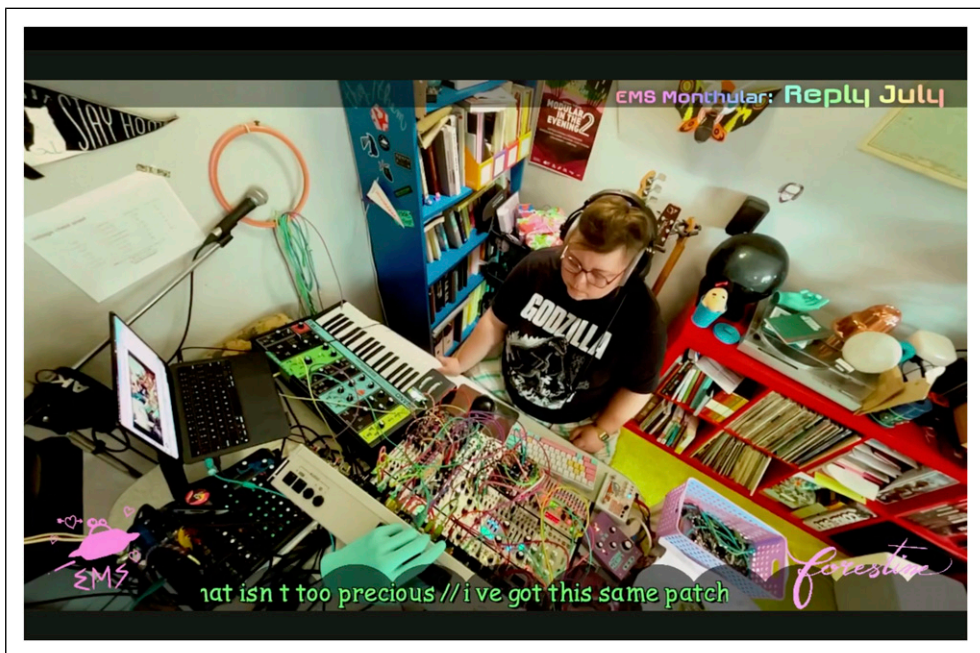


Figure 2. Screenshot from a forestine livestream for EMS.

keep the Discord and livestreaming platforms running smoothly—for a Zoom interview. The five who graciously accepted were schmooster (they/them, a UK-based musician and the originator of EMS), forestine (they/them, a nonbinary and chronically ill artist from the Canadian prairies and one of the first active users to become an admin), BbartokK (he/him, a Chilean artist based in Los Angeles), heckseven (he/him, a musician from the mountains of Colorado at 8000 feet), and Enthusiastic Electrons (he/him, a musician and budding module designer from Los Angeles, presently maintaining the YouTube stream archive, among other roles). There was such camaraderie amongst them that at times they were completing each other's sentences. And despite the numerous contributions all five have made over the years, none personally took credit for anything; they were quicker to credit others, including some who couldn't be in the interview but who had contributed significant things to EMS.

While some of the present EMS admins, especially heckseven, had a prior history with community organizing, for the most part there was no specific community model for EMS. It was built to accommodate specific needs and concerns: having a platform where anyone could perform, anytime, kind of like a radio station; facilitating conversations about these very complex technical objects that are often not well understood and where only a few users may really understand their quirks or features; and creating a safe space for free-ranging conversation among people who share a special niche interest not understood by others, that is, being excited about and consumed with the limitless possibilities that modular synthesis purports to present.

One unique performance genre that EMS developed they term a *stream train*. Stream trains began spontaneously in the first year of EMS; one member would be livestreaming their set and ask in the Discord chat if anyone wanted to hop on right after them. Early on these posed a logistical challenge since the software scripts and streamer-key-generation system hadn't yet been polished. As BbartokK recalls, an admin might pop into Discord to ask: "Hey, pinging different people! Are you ready? They got 1 minute to go. Are you here? Please respond." Then streamer number one would announce "Okay, I'm gonna hit stop in 5 seconds" so that streamer number two could try to precisely time the start of their stream—and so that the whole EMS channel wouldn't go down and need to be manually restarted by schmooster. But after a few users, particularly BenofBrown and Stogoshuffle, contributed custom code for the purpose, stream trains became a technically trivial way to pull off a wholly impromptu show lasting one, two, or more hours. Enthusiastic Electrons regards these as feeling "like the local bar or venue where it's open mic night, and you hang out. And you see your friends when you show up." This aspect might seem to align EMS' livestreams with those of music communities who similarly experimented with the format, for example, the #Verzuz rap battle concert series created by Timbaland and Swizz Beatz that aired in 2020 during the first COVID-19 lockdown in the US, where the livestreams became "representational spaces for live music experiences and intimate connection" in Black culture (Evans, 2001: 84). Unlike #Verzuz, EMS was not really a "representational space" as it quickly developed a distinctive aesthetics and vibe that was disconnected from external referents. EMS, additionally, was never conceived as a temporary stopgap to maintain an existing community during a time when live shows were impossible.

Whereas in Twitch gaming and political commentary channels a hype train serves mainly as a vehicle for encouraging donations to the streamer and for gifting subscriptions to other users (Jackson, 2020), EMS stream trains encourage collective participation in EMS members' live-streamed experiments, creations, ideas, and performances. In May 2021, some members planned an EMS-specific train for International Drone Day, figuring that a handful might participate. For the first time, they provided a theme (Brood X: Born to Drone) and cicada-themed overlay graphics that performers could use if they wished (Figure 3). In short order, artists had filled up eight and a half hours of performance slots; the stream train festival ended up running for over ten hours, with near constant chat percolating alongside the stream. Some performers used actual samples of cicadas

within their sets, while others riffed on their timbrally rich sound through analog synthesis and digital wavetable techniques.¹⁰

Seeing that semi-organized stream trains such as drone day filled a vital need—providing a space for modular artists of any skill level to experiment while garnering a critical mass of active chatters to support their efforts—EMS continued to experiment with the genre. As happened on the first drone day, after the scheduled sets had ended, a few artists tried to keep the collective energy going by hopping on to provide impromptu sets, what has become known as “picking up the caboose” of the stream train. The duration of stream trains also increased, with some of the drone days and EMS birthday party streams running continuously for more than 24 h.

EMS as a *doitocracy*

When I (naïvely) asked about the “democratic governance” and DIY ethos of EMS, and why they had chosen to organize things differently than every other modular society, admin heckseven corrected me: “I wouldn’t say we’re a democracy, because we’re not voting on things, although we have had some votes. But it’s more like group consensus. I’m not even sure that’s the right word, either. But it’s definitely more like the Anarchist cell.” Enthusiastic Electrons chimed in with his theory:

It feels like a *doitocracy*, where it’s like, if someone does it, then it gets done. Some things come up where someone’s like, “This would be a great idea.” And we’re like, “Great! It will not get done unless you do it, or if you know someone [who will do] it.” It seems like every time something has come up in this history of EMS, there’s someone who’s there, saying “I will do that,” just in terms of, like, writing a website, or generating a key system, an API, or whatever.



Figure 3. Flyer for EMS’ first stream train festival, Brood X: Born to Drone, May 29, 2021.

One example of the doocracy in practice is the creation of new EMS discord channels for local and regional members. As heckseven recalls about the formation of one, “BBartokK’s like, ‘hey, can we have an LA stream?’ Yeah, let’s do it, let’s promote the crap out of it, let’s go!” Subsequently, some of the LA members have gotten together in person to collectively stream from one location, and 28 additional channels focus on other regions (and in cases like the Mexico and Germany channels, languages other than English). EMS founder schmooster singled out this “going from the global to the local scope” and the possibility of “in person groupings” as standout, unexpected, positive EMS developments.

Another spontaneous creation was heckseven’s weekly talk radio-style hang called *What the Heck*. At the time he started it, there was already an active modular-related podcast ecosystem, but it was hosted by professional content creators and typically featured other professional creators or synth designers as guests. There was no medium for more casual modular discussion. As heckseven noted:

I just wanted to hang out and talk modular with some folks. I did one joint stream with Alex Barnett way back when and I was like, this is really fun. Can we have a weekly thing where we just hang out on camera, have some chatter? Anyone can join. The only requirement was that you have a microphone, and enough of an Internet connection to use the microphone. And we had, I don’t know, like 80-plus of those [episodes]. With lots of different folks on there... But that was an example of something that was like, it’d be fun to try this, let’s do it!

The anti-capitalist ethos of founder schmooster became an anti-capitalist ethos for the whole community. EMS do not monetize their Twitch or YouTube channels; when they do produce merch (especially t-shirts) they sell it at cost since “the point of it is so that people can have shirts,” according to Enthusiastic Electrons. As heckseven put it, although everyone works to promote larger events on various channels, the group doesn’t even especially care to arbitrarily grow the viewership, since “we want all the people *who want to be involved*” (emphasis in original).

Maintaining a doocracy

EMS, in 2026, stands out among the several dozen online modular communities in being the most identity-diverse space, although there is scant specific messaging about this on the streams, Discord, or website. Since members are not required to show their body in the livestream frame—they may focus the camera only on their modular, or use motion graphics instead of depicting themselves and their instrument—this has permitted synthesists to stream without sharing any aspect of their sociocultural identity, if they wish. (Nearly all streamers show their modular setup, though). Concerning how this all came to be, forestine suggested that:

I do think it helped that, even from the beginning, it was fairly gender and identity diverse. I found a lot of places like reddit and forums to be really bro-y. It just feels very... the specific kind of guy like... no offense, but like, you know.

Since prospective new members were likely to come from hierarchical online communities, and may have never directly participated in DIY/DIT communities of any sort, the EMS admins soon realized that every user’s first experiences in the community were crucial. As Enthusiastic Electrons put it:

At EMS we don't have an automatic, robot-generated welcome template. Whenever anyone new shows up, whoever is around [will give] a personal, "Hey, welcome to EMS." And also some question like, "what's your favorite synth?" to see if they're a bot or not. There's a lot of constant work that goes on to keep it a clean server where it's, you know, respectful humans.

heckseven was explicit about the intent behind this:

If you're going to be a welcoming community that gives a shit about diversity, equity, inclusion, and people not being assholes, you have to set the tone right away. I don't know if it was serendipitous, or the type of person that schmooster is, and attracted, initially. But the folks that we got, including the folks in this room, really give a shit about all of that, and that meant active discussions publicly in our public channels about: What kind of rules do we want? What kind of people do we want to be? That kind of stuff resulted in a community that eventually grew into something that's very welcoming.

Along the same theme, although 17 users are marked in Discord as "admins" and have moderator privileges on Twitch, none regard themselves as a "leader." forestine once characterized the collective as "a bunch of raccoons in a trench coat." And BbartokK emphasized a decentralized structure when talking about the generation and adoption of new rules:

From the beginning... any types of these questions were discussed openly and were listened to openly. There wasn't a behind the scenes of "oh, now there's this rule, we don't know why, but we have to follow it." It was always out in the open, and I think that's like the open-source mentality that schmooster has always had.

And as heckseven pointed out, the admins enact an ethics of care regarding other members and their wellbeing, especially concerning one common problem: member burnout. However, being that EMS is tethered to modular gear cultures, rather than representing a problem of various users' interactions with each other, burnout often relates to the individual relations that a member has with their gear:

In a lot of communities, professions, hobbies, folks [can] go too hard on something, whether that's streaming or modular. So we do keep an eye out and try to support anyone that's feeling burnout, or some sense of guilt for one thing or another. Like there's folks that like tend to buy modules back-to-back, and they start feeling guilty when they haven't used them all. We just try to like, look out for each other like, and create a healthy spot. (heckseven)

If burnout is not effectively addressed, it can lead to a member selling all their modular gear (a "fire sale") and quitting the community outright.

Discussion

How does the EMS case study compare with previously researched sociotechnical and music collective practices? While Joseph Ugoretz attributed the rise of doitocracies to the open-source software movement, where "whoever does the most, gets the most credit" (2013), in EMS there never has been an obvious goal of accruing credit. Other than the transparent governance that BbartokK mentioned there is no mandated open-source ethos; beyond the module schematics designed by some of the members participating in the DIY channel, or code written to hack digital modules, little else *could* be open sourced. Therefore, this represents a different kind of doitocracy.

There are some superficial similarities to the “entanglements between individuals and material space” of underground DIY noise music scenes (Woods and Ortega, 2025) or the “underground resistance” of DIY punk scenes (Guerra, 2018). However, the central role of synthesizers and other technological objects as full-fledged actors, the conspicuous consumption of these low-ticket luxury goods, and the considerably diminished role of music fandom in the milieu, demarcates this doitocracy as a contrasting case study. Prior work has examined the central role of platform-specific social interaction in the formation of genre-specific local music scenes, as was the case with the Budapest lo-fi scene that moved from Tumblr blogs to in-person events (Barna, 2019). But the result for EMS was not a local music scene, but a gear culture that was international, not local, in scope—and one that traverses several platforms.

This kind of doitocracy relates to, yet differs from, the *produsage* concept as outlined by Axel Bruns (2010: 3). Bruns was examining a new kind of participatory culture around blogs, mashups, and early social media, and identified four “universal principles” for produsage:

- (1) open participation and communal evaluation
- (2) fluid heterarchy and an ad hoc meritocracy
- (3) unfinished artefacts and a continuing process
- (4) common property yet individual rewards

In other words, the produsage moniker applies ideally to systems for creating, producing, evaluating and sharing “content” in the form of works; a produsage system stands in tension with long-standing intellectual property concepts of ownership and authorship. But for EMS, there is neither a specific shared aim to create works nor any sense that any works that might exist are either finished or unfinished. While open-source content is optional, some EMS broadcasts and Discord discussions have resulted in commercially released albums or synthesizer modules. Most importantly, while participation is “open” to anyone regardless of skill, communal evaluation is conspicuously avoided: the streams and Discord are not places for critiquing other members’ performances, but rather intended for encouraging everyone along their modular journey.

Perhaps the closest parallel to EMS, despite the different platforms, activities, technologies, and purposes, is the low-power FM and pirate radio movement studied by Christina Dunbar-Hester. EMS’ stream trains, though formally very different, in the sense of all-hands-on-deck collective action have similarities with the “barnraisings” for creating new LPFM stations (Dunbar-Hester, 2014: 19–26). The doing/making aspect of the doitocracy, which manifested in user contributed code to improve the streaming platform’s functionality, and in the mass participation on the stream platform as performers, means that gear culture members arguably foster *geek identities* (ibid.:31–37) and *technical identities* (ibid.:120–21) akin to those of radio activists. In both cases, the main work represents a political intervention in for-profit media landscapes even though the platform is not used to promote any one political ideology and lacks any stated political agenda (ibid.:186).

To be clear, the doitocracy of EMS arose in opposition to certain toxic spaces that dominated synth sociability in 2020. In formal terms, it represents the historically bounded results of a strategy for counteracting the lack of participatory spaces specific to one hobby. I do not provide a conclusive definition of doitocracy, since other doitocracies might differ in form and values. I also do not assert that EMS’ strategy would work equally well in other settings. That said, I hope to have conveyed the sophistication of the theory-building and world-building work that amateur synthesis enthusiasts perform when they craft communities around thousands of

geographically distant human-gear encounters. I also hope to have shown the considerable role of fetishized material technologies, and human-material intra-actions, in facets of online social life ranging from musical performances to collaborative design projects to platform instance governance.

Conclusion

Gear cultures are post-scene social formations that, so far, have been viewed as hyper-capitalist hegemonic masculinities coalescing around specific categories of fetishized gear. The doitracy of EMS appears to represent a new development: a community that deliberately deemphasizes several central formal aspects of preceding gear culture instances. When gear culture participation entails individuals jockeying for social status and cultural capital through their practices of conspicuous consumption, this benefits those who enter the gear culture with the most disposable money. More often than not this arises from intergenerational wealth (Douglas and Isherwood, 1996: 114). However, as is evident in the EMS case study, when gear culture activities are instead organized around DIY/DIT/doitracy practices, and when the community abandons the top-down hierarchies of the YouTube Content Creator and regional society communities, the incentives are instead *to do*, and to inspire others to do. The result has the potential to be a collective entity that serves as an umbrella for the individual projects of many, whether those projects are musical performances, product design, coding, or leading a group discussion. Importantly, by allowing its pseudonymous members to represent themselves however they wish, EMS came to lack a hegemonic gender ideology and became, in a practical sense, more inclusive.

The platform-jumping nature of a doitracy such as EMS poses challenges and opportunities for research into online communities. The fluid mobility of internet sociality between YouTube, Twitch, Discord, and message fora, revealed in the reflections of the EMS admins, reaffirms Daniel Miller's argument that "to understand social media we need to focus instead upon *content*, which often migrates and switches easily between entirely different platforms almost regardless of their properties" (2016: 1). The "content" here, though, is irreducible to "music" and "chat," since the central organizing feature of most interaction is the irreducible materiality of synthesis gear. Gear, here, is not "represented" but rather *performed* by performers. Considering the numerous other gear cultures where fetishized, crafted objects play a central role in online social interaction, there is a palpable need for analytical frameworks that move beyond representationalism and that analyze sociality via the material performances that help make community possible.

Acknowledgments

I thank the five Earth Modular Society admins—forestine, schmooster, heckseven, BbartokK, and Enthusiastic Electrons—for their generosity in participating in this research.

ORCID iD

Eliot Bates  <https://orcid.org/0000-0003-3012-1949>

Ethical considerations

Ethics approval was obtained from the CUNY Graduate Center prior to conducting the research, and informed consent was obtained from all research participants.

Consent for publication

All participants had the opportunity to review and modify the statements attributed to them, as well as the general statements made about EMS, prior to publication.

Funding

The author disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: Support for this research was provided by a PSC-CUNY Research Award from The City University of New York (67620-00 55).

Declaration of conflicting interests

The author declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Notes

1. Regarding data ethics, since the Discord server and the livestream comments appear to be regarded by some participants as semi-private spaces, I chose not to quote from them or discuss content that could identify specific users other than those whom I formally interviewed.
2. <https://www.youtube.com/@EarthModularSociety/playlists> [accessed 10 Aug 2025].
3. See also Ted Gordon's critique of the analog revival narrative, which he argues constructs an anachronistic and embellished portrayal of the motivations of some of the synth designers at the time (Gordon 2024).
4. In modular, the pitch of a sound is determined by a control voltage, which could be produced by a keyboard (where every key produces a different voltage), or by any other changing voltage source. Generative patches creatively play with different voltage sources to create never-repeating strings of notes that are inherently unpredictable, but still sound "musical."
5. Interview with Earth Modular Society admins BbartokK, Enthusiastic Electrons, forestine, heckseven and schmooster. April 3, 2025. Zoom. All quotes from these five individuals, in this article, come from this interview, unless indicated otherwise.
6. The author has conducted an informal, small-scale analysis of digital photography gear cultures as a control to the present study, and works by Herbst and Menze (2021) and Brounley (2022) provide insight into the consumer cultures of guitar pedals and guitar amps, but only partial and local accounts of the rest have been published by other scholars.
7. Recording studio and modular synthesis gear cultures came about just when software plugins and software modular instruments were replacing hardware gear in studio contexts, and photography gear cultures appear to have arisen partly as a refutation of smartphone photography.
8. While it is beyond the scope of the present study, the many Discord servers run by small instrument design firms, especially Winterbloom, This is Not Rocket Science, and Plinky/Making Sound Machines, serve as important nodes for synth design pedagogy. Beyond the discussion of the company's own products, users also discuss circuit design, digital signal processing code, and the logistics of product manufacturing.
9. As one indicator of the gender composition of these spaces, not once during nearly two decades of online platform discussion have I seen any discussion of how to convince "the husband" about the necessity of a gear purchase.
10. For example, in my duo performance with Arseni Troitski as Manifestoon Platoon, both of us applied granular synth processing to Brood X cicada samples I had field-recorded days before at Hopewell, NJ. That set, and the full playlist, can be heard/watched at <https://www.youtube.com/watch?v=thchXNoQ2Io>.

References

- Barna E (2019) Birth of an underground music scene? Creative networks and (digital) DIY technologies in a Hungarian context. In: Bennett A and Guerra P (eds) *DIY Cultures and Underground Music Scenes*. Routledge, pp. 171–182.
- Bates E (2022) La musique dans les cultures tech. La synthèse modulaire en temps de Covid-19. *Cahiers d'Ethnomusicologie* 35: 225–243.
- Bates E (2026a) Gear culture methods: materialising intra-actions and fetish relations. *Science and Technology Studies*, Epub ahead of print 2026.
- Bates E (2026b) *Patching pandas to pressure points: the paradoxes and promises of modular synthesis*.
- Bates E and Bennett S (2022) Look at all those big knobs! Online audio technology discourse and sexy gear fetishes. *Convergence: The International Journal of Research Into New Media Technologies* 28(5): 1241–1259.
- Bates E and Bennett S (2025) *Gear: Cultures of Audio and Music Technologies*. The MIT Press.
- Baudrillard J (1998) *The Consumer Society: Myths and Structures*. Sage.
- Beer D (2013) Genre, boundary drawing and the classificatory imagination. *Cultural Sociology* 7(2): 145–160.
- Brounley DM (2022) *That Sound in Your Head: Guitar Tone Values in the Entrepreneurial Age*. Ph.D. Dissertation. Stony Brook University.
- Brown W (2006) American nightmare: neoliberalism, neoconservatism, and de-democratization. *Political Theory* 34(6): 690–714.
- Bruns A (2010) Distributed creativity: filesharing and produsage. In: Sonvilla-Weiss S (ed) *Mashup Cultures*. Springer, pp. 24–37.
- Davies SR (2017) Characterizing hacking: Mundane engagement in US hacker and makerspaces. *Science, Technology & Human Values* 43(2): 171–197.
- DeNora T (2000) *Music in Everyday Life*. Cambridge University Press.
- Douglas M and Isherwood B (1996) *The World of Goods: Towards an Anthropology of Consumption*. 2nd edition. Routledge.
- Dunbar-Hester C (2014) *Low Power to the People: Pirates, Protest, and Politics in FM Radio Activism*. The MIT Press.
- Evans J (2001) Old hits veruz new technology: how a pandemic ushered legacy artists into monetizing the clout economy. In: Miller NP (ed) *Sustaining Black Music and Culture During COVID-19: #Veruz and Club Quarantine*. Lexington Books, pp. 79–94.
- Fisher M (2009) *Capitalist Realism: Is There No Alternative?* Zero Books.
- Ging D (2017) Alphas, betas, and incels: theorizing the masculinities of the manosphere. *Men and Masculinities* 22(4): 638–657.
- Gordon T (2024) The buchla music easel: from cyberculture to market culture. In: Teboul E, Kitzmann A and Engström E (eds) *Modular Synthesis: Patching Machines and People*. Routledge, pp. 49–80.
- Guerra P (2018) Raw power: punk, DIY and underground cultures as spaces of resistance in contemporary Portugal. *Cultural Sociology* 12(2): 241–259.
- Henry T (2008) *The Electronic Drum Cookbook*. lulu.com: Magic Smoke Electronics.
- Hepp A and Schmitz A (2022) The limits of the maker ideology: local makerspaces, experimental practices, and COVID-19. *Continuum* 36(2): 199–213.
- Herbst J-P and Menze J (2021) *Gear Acquisition Syndrome: Consumption of Instruments and Technology in Popular Music*. University of Huddersfield Press.
- Hutchins B (1972–2004) *Electronotes*.
- Jackson NJ (2020) Understanding memetic media and collective identity through streamer persona on Twitch.tv. *Persona Studies* 6(2): 69–87.

- Kitzmann A and Engström E (2024) Introduction. In: Teboul E, Kitzmann A and Engström E (eds) *Modular Synthesis: Patching Machines and People*. Routledge, pp. 1–17.
- McRobbie A (1980) Settling accounts with subcultures: a feminist critique. *Screen Education* 34: 37–49.
- Miller D (2016) *Social Media in an English Village*. UCL Press.
- Morgan F (2017) Delian modes: listening for Delia Derbyshire in histories of electronic dance music. *Dancecult: Journal of Electronic Dance Music Culture* 9(1): 9–27.
- Navs (2016) Basic electricity: an appeal for a greater understanding of rudimentary modular functions. *eContact!* 17(4). Available at: https://econtact.ca/17_4/navs_basicelectricity.html.
- Nelson A, Anthony C and Tripsas M (2023) “If I could turn back time”: occupational dynamics, technology trajectories, and the reemergence of the analog music synthesizer. *Administrative Science Quarterly* 68(2): 551–599.
- O’Grady P (2021) Rethinking criticism about lossy compression: sound fidelity, large-scale production and audio capital in pop music. *Convergence* 27(4): 1075–1091.
- Pels P (2023) *The Spirit of Matter: Religion, Modernity, and the Power of Objects*. Berghahn Books.
- Pinch T and Trocco F (2002) *Analog Days: The Invention and Impact of the Moog Synthesizer*. Harvard University Press.
- Rogers J (2020) ‘She made music jump into 3D’: Wendy Carlos, the reclusive synth genius. *The Guardian*, November 11. <https://www.theguardian.com/music/2020/nov/11/she-made-music-jump-into-3d-wendy-carlos-the-reclusive-synth-genius>
- Schedel M (2017) Electronic music and the studio. In: Collins N and D’Escrivan J (eds) *The Cambridge Companion to Electronic Music*. Cambridge University Press, pp. 25–37.
- St John G (2004) Techno millennium: dance, ecology and future primitives. In: St John G (ed) *Rave Culture and Religion*. Routledge, pp. 213–235.
- Straw W (2001) Scenes and sensibilities. *Public* 22–23: 245–257.
- Teboul E (2024) Preface. In: Teboul E, Kitzmann A and Engström E (eds) *Modular Synthesis: Patching Machines and People*. Routledge, pp. 18–48.
- Troitski A and Bates E (2024) Grid culture. In: Teboul E, Kitzmann A and Engström E (eds) *Modular Synthesis: Patching Machines and People*. Routledge, pp. 205–224.
- Ugoretz J (2013) The emerging doitocracy. In: *Prestidigitation*. Available at: <https://prestidigitation.commons.gc.cuny.edu/2013/02/10/the-emerging-doitocracy/> (accessed 31 December 2025).
- Veblen T (1899) *The Theory of the Leisure Class: An Economic Study in the Evolution of Institutions*. MacMillan Company.
- Wasielewski A (2017) *Made in Brooklyn: Artists, Hipsters, Makers, Gentrifiers*. Zero Books.
- Wilson R (2013) *Make: Analog Synthesizers*. Maker Media.
- Woods PJ and Ortega Y (2025) Scenes of entanglement: towards a posthuman understanding of the transglobal noise music scene. *DIY, Alternative Cultures & Society* 3(1): 3–17.